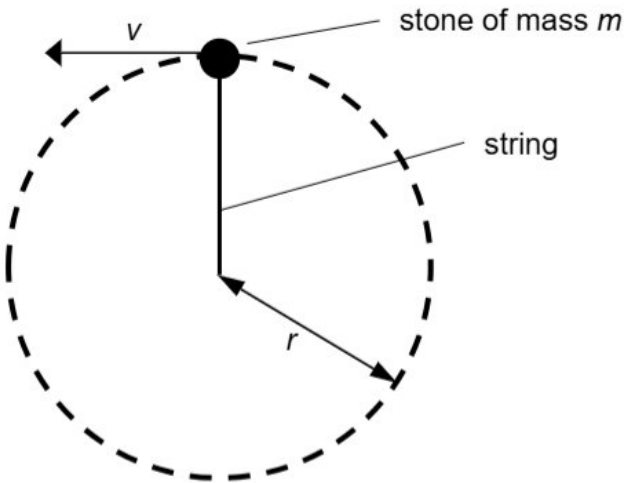
8 A stone of mass m is attached to a string. The stone is made to rotate in a vertical circle of radius r, as shown.



At the point where the stone is vertically above the centre of the circle, the speed of the stone is v. Which of the following expressions gives the tension in the string?

A $mg - \frac{mv^2}{r}$ **B** $\frac{mv^2}{r}$ **C** $\frac{mv^2}{r} - mg$ **D** $\frac{mv^2}{r} + mg$